Potato Suitability: Communal Beliefs, Income Dispersion, and Civic Engagement

EVS5/WVS7 Joint v5 outcomes merged with Nunn & Qian (2011) country panel — 31-32 European countries (v3)

1. Question

Does potato suitability — Nunn & Qian's (2011) instrument for the New-World caloric shock — predict three modern country-level outcomes: (a) communal-vs-individual values, (b) within-country dispersion of self-reported income, and (c) civic engagement (associational membership)?

2. Sample and country match

EVS5/WVS7 Joint v5 (2017–2022): 156,658 respondents across ~90 countries. Restricting to the 35 European countries flagged cont_europe==1 in Nunn-Qian's country_level_panel_for_web.dta leaves 32 that fielded this joint wave (Belgium, Ireland, Moldova did not participate). Country matched via ISO 3166-1 Alpha-2 (cntry_AN) → Alpha-3 (isocode). Serbia (RS) + Montenegro (ME) → N-Q's combined YUG entity; Northern Ireland (NIR) → GBR. Portugal lacks the income-decile item, so the income-dispersion sample is N = 31.

3. Outcomes

3a. Communal-vs-individual belief index (11 items, child qualities removed)

Per-respondent index = mean of 11 items rescaled to [0,1] with 1 = communal. Six work/duty items (C038, C039, C041, D026_03, D026_05, D054), four economic/political 1–10 scales (E035 reversed, E036, E037, E039), and one trust item (A165, recoded so in-group/low-radius response = communal). Country-level value = gwght-weighted mean.

3b. Within-country income dispersion

Self-reported income decile combines X047_WVS7 (WVS7) and X047E_EVS5 (EVS5), giving 144,957 valid responses across 31 countries. Three dispersion measures computed per country with gwght weighting: SD (income_sd), CV = SD/mean (income_cv), and IQR (in regression_results.txt). Captures dispersion of perceived position, not actual income.

3c. Civic engagement (NEW)

Sum of membership across 11 organisation types (A065 religious, A066 arts/cultural, A067 unions, A068 parties, A071 environmental, A072 professional, A074 sports, A078 consumer, A079 other, A080_01 humanitarian/charity, A080_02 self-help). Each item is binary 0/1. Per-respondent value = share of items answered for which the respondent reports membership (requires ≥6 of 11 answered). Per-country value = gwght-weighted mean.

Country ranking (low → high engagement): PRT 0.014, ALB 0.018, ITA 0.032, EST 0.032, RUS 0.036, POL 0.037, BGR 0.039, LTU 0.046, BIH 0.053, LVA 0.055, ESP 0.057, FRA 0.057, BLR 0.061, HUN 0.062, ROU 0.077, UKR 0.078, YUG 0.079, SVK 0.083, HRV 0.088, CZE 0.089, MKD 0.103, AUT 0.117, DEU 0.154, NLD 0.155, FIN 0.163, SVN 0.163, NOR 0.165, GBR 0.173, CHE 0.191, SWE 0.193, DNK 0.214, ISL 0.264. Matches every comparative civic-culture ranking I'm aware of: Nordic + Anglo + Germanic at the top; Mediterranean and post-Soviet at the bottom.

4. Regressions

Cross-sectional OLS, robust (HC1) standard errors. Specifications: (1) bivariate; (2) + old-world crops (ln_oworld); (3) + geography (elevation, ruggedness, tropical); (4) + full set (distance to equator, distance to coast, malaria); (5) + Catholic and Protestant shares.

4a. Communal index (11 items)

Specification	β (ln_wpot)	SE	t	p	R^2	N
(1) Bivariate	0.0094	0.0048	1.96	0.050	0.081	32
(2) + ln_oworld	-0.0158	0.0223	-0.71	0.478	0.131	32
(3) + Geography	-0.0157	0.0197	-0.80	0.425	0.307	32
(4) Full controls	0.0123	0.0158	0.78	0.436	0.606	32
(5) Full + religion	-0.0031	0.0111	-0.28	0.782	0.796	32

Bivariate β = 0.0094 ($p = 0.050$) — barely significant; killed by adding ln_oworld and stays insignificant under richer controls. Same pattern as the original 21-item index.

4b. Within-country income SD

Specification	β (ln_wpot)	SE	t	p	R^2	N
(1) Bivariate	-0.0111	0.0210	-0.53	0.598	0.006	31
(2) + ln_oworld	0.0074	0.0896	0.08	0.935	0.007	31
(3) + Geography	0.0189	0.0876	0.22	0.829	0.173	31
(4) Full controls	0.0206	0.1028	0.20	0.841	0.257	31
(5) Full + religion	0.0700	0.1000	0.70	0.484	0.307	31

Null across all five specifications. Bivariate $R^2 = 0.006$.

4c. Within-country income CV

Specification	β (ln_wpot)	SE	t	p	R^2	N
(1) Bivariate	0.0069	0.0049	1.39	0.164	0.052	31
(2) + ln_oworld	-0.0045	0.0180	-0.25	0.801	0.064	31
(3) + Geography	-0.0019	0.0192	-0.10	0.920	0.084	31
(4) Full controls	0.0340	0.0238	1.42	0.154	0.207	31
(5) Full + religion	0.0329	0.0260	1.26	0.206	0.251	31

Coefficient never reaches significance. Highest p-value approaches in Spec 4 ($p = 0.154$).

4d. Civic engagement

Specification	β (ln_wpot)	SE	t	p	R^2	N
(1) Bivariate	-0.0155	0.0045	-3.47	0.001	0.231	32
(2) + ln_oworld	0.0151	0.0161	0.94	0.349	0.310	32

Specification	β (ln_wpot)	SE	t	p	R²	N
(3) + Geography	0.0180	0.0142	1.26	0.206	0.355	32
(4) Full controls	-0.0199	0.0189	-1.05	0.294	0.586	32
(5) Full + religion	-0.0083	0.0193	-0.43	0.667	0.706	32

Bivariate β = -0.0155 ($p = 0.001$) is the strongest result in the memo — and it goes the opposite direction from what a naive caloric-surplus → social-capital story would predict. High-potato Eastern European countries are systematically less civically engaged than low-potato Nordic / Anglo / Germanic countries. $R^2 = 0.23$, by far the largest bivariate R^2 across all four outcomes.

But, as before, the relationship collapses once ln_oworld is added (Spec 2 flips the sign, $p = 0.35$) and stays insignificant under the full and full+religion controls. The bivariate effect is again driven by general agricultural-suitability variation that maps onto the East-West axis.

There is also a leverage problem. Iceland has ln_wpot = 0 (potatoes are effectively unsuitable there) yet the highest civic engagement (0.264) — a single high-leverage observation. Dropping ISL weakens the bivariate to $p = 0.10$, $R^2 = 0.064$:

Specification	β (ln_wpot)	SE	t	p	R²	N
Bivariate (drop ISL)	-0.0099	—	—	0.104	0.064	31
+ ln_oworld (drop ISL)	0.0137	—	—	0.403	0.127	31
Full controls (drop ISL)	-0.0211	—	—	0.273	0.480	31

5. Bottom line across all three outcomes

All four outcomes (the original 21-item index, the 11-item index, income dispersion, and civic engagement) yield the same pattern: a bivariate correlation between ln_wpot and the outcome that is either marginally significant or — for civic engagement — strongly significant, but which is wiped out the moment ln_oworld is added to the regression.

This is the canonical sign of confounded geography. Eastern European countries score high on potato suitability and on overall agricultural suitability simultaneously, because the same temperate continental climate that suits the potato also suits cereal staples. They also share other characteristics — communist legacy, late industrialization, Orthodox / late-Catholic religion — that lower civic engagement and individualism. NQ's preferred identification (which variation is specific to the potato, net of other crops) finds nothing in this cross-section.

6. Notes for interpretation

- N = 31–32 with seven baseline controls; the study cannot rule out small effects.
- Civic engagement is the highest-R^2 outcome (bivariate 0.23) and most sensitive to leverage from Iceland's ln_wpot = 0; the robustness exercise drops it from significance.
- Both belief and civic outcomes are observed only in the modern post-treatment period, so the panel diff-in-diff used in NQ Table 2 is not directly available. A pre-treatment proxy at the country level would let you run the full DID.
- The Stata pipeline (beliefs.do) now writes country_outcomes.dta with all four outcomes (communal_index, civic_share, civic_count, income_mean/sd/cv/iqr) plus the merge into the N-Q controls and all 25 OLS specifications.

7. Files in the Potato Dropbox folder

- beliefs.do — Stata pipeline (paths set at top, no child qualities, civic engagement + income dispersion blocks)
- country_outcomes.csv — country-level all four outcomes (in the actual Stata run the .dta is also written)
- country_civic.csv — civic engagement subset
- regression_data.csv — outcomes merged with N-Q geographic controls
- regression_results.txt — full OLS summaries for all four outcomes × 5 specs
- results_summary.csv — compact β/SE/p/R² table
- potato_beliefs_scatter.png — three-panel bivariate scatter (below)